# Superdeterminism lives below every resolution; a floor gives it a lifetime

**Result, 2026-09-29 (Claude; written derivation, refereed by Fable with REFINE, corrections applied below).** Classical mechanics is deterministic, so the settings of a Bell test are functions of the initial data; the statistical independence of settings and hidden variables that Bell's theorem needs is an assumption of typicality on those data, taken with the Liouville measure. Superdeterminism is the choice of atypical data. The proposition below shows where such data must hide when the setting device is chaotic: in structure of the device's initial ensemble finer than any fixed phase-space resolution. For a preparation whose device part is resolved at scale $\delta$, the covariance between any bounded hidden variable $f$ and a setting chosen $n$ steps later obeys

$$|\mathrm{Cov}_\rho(f, g\circ T^n)| \le 2C_T\,K\,\sup|f|\,\|g\|_{\mathrm{Lip}}\,\delta^{-(d+1)}\theta^n,$$

so the correlation is below $\epsilon$ after

$$n_*(\delta) = \frac{(d+1)\ln(1/\delta) + \ln(2C_T K \sup|f|\,\|g\|_{\mathrm{Lip}}/\epsilon)}{\ln(1/\theta)}$$

steps. Classical mechanics allows every $\delta > 0$, and $n_* \to \infty$ as $\delta \to 0$: a conspiracy can be stored in ever finer structure for as long as needed. If a floor $h$ on phase-space area per canonical pair constrains the prepared state, then $\ln(1/\delta) = \frac{1}{2}\ln(A_0/h)$ with $A_0$ the device's action scale per pair, and the conspiracy has a finite lifetime $n_* \approx \frac{d+1}{2\ln(1/\theta)}\ln(A_0/h)$, the logarithmic structure of the Ehrenfest time of quantum chaos (Berman and Zaslavsky 1978; metadata). The dependence of hidden-variable events on setting events then decays exponentially after an onset of order $2n_*$ (Remark in §1), and relaxed Bell inequalities bound local deterministic models that respect the floor, under the hypotheses stated there. The limit $h \to 0$ that Newton takes is exactly what reopens the loophole.

The user's remark that prompted this (2026-09-28/29): classical mechanics is accidentally superdeterministic too, and the programme holds that it has a mistake somewhere. The link is that typicality rests on the Liouville measure, whose unit is an action per degree of freedom, the same object whose unlimited refinement the fifth-postulate note identifies as Newton's joint determinacy.

## 1. Setting and the proposition

The phase space is a product $M_{\mathrm{sys}} \times M$: the measured system, a probability space with reference measure $\nu$ and otherwise arbitrary, and the setting device, a compact **connected** $d$-dimensional manifold with normalized volume $\mu$. The dynamics is $\mathrm{id} \times T$, with $T$ a $C^2$ volume-preserving Anosov diffeomorphism; only the device needs chaos. By Anosov's theorem such a $T$ is ergodic for $\mu$, hence transitive, hence topologically mixing because $M$ is connected, and $\mu$ is its equilibrium state. Bowen's theory then gives exponential decay of correlations for Hölder, hence Lipschitz, observables (Bowen 1975; metadata, theorem numbers still to be checked at passage level): with $\|F\|_{\mathrm{Lip}} = \sup|F| + \mathrm{Lip}(F)$,

$$\Big|\int F\,(G\circ T^n)\,d\mu - \int F\,d\mu \int G\,d\mu\Big| \le C_T \|F\|_{\mathrm{Lip}}\|G\|_{\mathrm{Lip}}\,\theta^n, \qquad \theta < 1,$$

where $\theta$ is the transfer-operator gap rather than a Lyapunov exponent.

A preparation is a probability density $\rho(x_1, x)$ on $M_{\mathrm{sys}} \times M$ with respect to $\nu \otimes \mu$. The hidden variable is $\lambda = f(x_1)$ with $f$ merely bounded and measurable, and the setting is $s = g(T^n x)$ with $g$ Lipschitz; $\lambda$ is fixed at time 0 and the setting is read $n$ steps later. The preparation is **resolved at scale** $\delta$ in the device if $\int \|\rho(x_1,\cdot)\|_{\mathrm{Lip}}\,d\nu(x_1) \le K\delta^{-(d+1)}$, i.e. the device's density

conditional on the system is Lipschitz at scale $\delta$ on average. For example, a conditional density smoothed by a mollifier $\phi_\delta$ of width $\delta \le 1$ has height at most $\delta^{-d}\sup\phi$ and slope at most $\delta^{-(d+1)}\mathrm{Lip}\,\phi$, so $K = \|\phi\|_{\mathrm{Lip}}$.

**Proposition.** For a preparation resolved at scale $\delta$ in the device, $|\mathrm{Cov}_\rho(f, g\circ T^n)| \le 2C_T K \sup|f|\,\|g\|_{\mathrm{Lip}}\delta^{-(d+1)}\theta^n$.

*Proof.* For each $x_1$ apply the decay estimate with $F = \rho(x_1,\cdot)$: $\int \rho(x_1,x)g(T^n x)\,d\mu = \rho_1(x_1)\int g\,d\mu + E(x_1)$, with $\rho_1$ the system marginal and $|E(x_1)| \le C_T\|\rho(x_1,\cdot)\|_{\mathrm{Lip}}\|g\|_{\mathrm{Lip}}\theta^n$. Then $E_\rho[f\,g\circ T^n] = E_\rho[f]\int g + \int fE\,d\nu$ and $E_\rho[g\circ T^n] = \int g + \int E\,d\nu$, so $\mathrm{Cov}_\rho = \int fE\,d\nu - E_\rho[f]\int E\,d\nu$, bounded by $2\sup|f|\int|E|\,d\nu$. □

The product form frees the hidden variable from any regularity: a superdeterminist may choose $f$ at will, and the conspiracy must then sit in the device's conditional density.

**Remark (from covariances to measurement dependence).** Bell's theorem uses the laws of $\lambda$ conditional on the settings. Hall measures their dependence by the $L^1$ (total-variation) distance between these conditional laws and gives relaxed Bell inequalities in its terms, together with a deterministic model of the singlet that needs only a partial dependence (Hall 2010); Barrett and Gisin quantify the missing independence in information-theoretic form (Barrett and Gisin 2011). Both: Crossref metadata verified 2026-09-29, content from memory. Let a setting event $A \subset M$ have a piecewise smooth boundary whose $w$-neighbourhood has $\mu$-measure at most $c_A w$, and let $B$ be any measurable hidden-variable event. Replace $1_A$ by a function with values in $[0,1]$ and Lipschitz constant $1/w$ that differs from it only in that neighbourhood, so its norm is at most $2/w$ for $w \le 1$. The device marginal has $\sup\rho_{\mathrm{dev}} \le K\delta^{-(d+1)}$, and since $T$ preserves $\mu$ the layer costs at most $K\delta^{-(d+1)}c_A w$ in $E_\rho[1_B\,1_A\circ T^n]$ and in the product of the means. Hence

$$\sup_B |\mathrm{Cov}_\rho(1_B, 1_A\circ T^n)| \le K\delta^{-(d+1)}\big[4C_T\theta^n w^{-1} + 2c_A w\big],$$

and $w = \theta^{n/2}$ gives $\varepsilon_n = K\delta^{-(d+1)}(4C_T + 2c_A)\theta^{n/2}$. For setting cells $A_j$ with probabilities $p_j = P_\rho(T^n x \in A_j)$, which tend to $\mu(A_j)$ at the same rate, the conditional laws of $\lambda$ satisfy $\sup_B |P(B\mid A_j) - P(B\mid A_k)| \le \varepsilon_n(1/p_j + 1/p_k)$. The onset for events is twice the Lipschitz $n_*$, with the same logarithm of $A_0/h$. The conclusion that relaxed Bell inequalities then exclude the singlet correlations holds under three stated hypotheses: (a) the delay $n$ between fixing $\lambda$ and reading the settings exceeds this onset; (b) setting cells have regular boundaries (hidden-variable events need none); (c) both wings' settings come from one Anosov device, or from two, whose product is again Anosov on a connected manifold.

**Scope of the floor.** The floor is imposed on the preparation; the dynamics is Newton's. The pushforward of $\rho$ stretches along unstable directions, so its Lipschitz norm grows like $\theta^{-n}$ and violates the floor at about the same $n_*$. A floor enforced at every step, by a coarse-graining map, could only add decorrelation; that is left unproved. Read quantum mechanically, Husimi functions are resolved at $\delta = (\hbar/A_0)^{1/2}$ in this sense, but classical evolution of observables tracks the quantum one only up to the Ehrenfest time, i.e. up to $n_*$. The floor meant here is the programme's classical floor, and the Ehrenfest match is formal. The translation $\ln(1/\delta) = \frac{1}{2}\ln(A_0/h)$ assumes the cell is split isotropically between $q$ and $p$ after both are normalized by $A_0^{1/2}$, with $d = 2N$ for $N$ canonical pairs.

## 2. What it says, and about whom

- **Classical mechanics.** The laws leave $\delta$ free. For a chaotic setting device, independence is therefore an assumption about the data, and the superdeterministic option is a device ensemble with structure at the scale $\theta^n$ or finer along the stable directions, correlated with $\lambda$. This is the precise sense of "accidentally superdeterministic"; for an integrable device, coarse data can carry a correlation forever, and the statement is proved only in the chaotic case.

- **A floor.** The proposition needs the floor to constrain the prepared physical state (its phase-space density), beyond constraining records. Under that premise, correlations of a chaotic setting device with a hidden variable decay after a time logarithmic in $A_0/h$, the Ehrenfest-type scale at which classical and quantum chaos part.
- **'t Hooft's automaton** lies outside the proposition. Its ontic states are exact cells and its correlations are carried by exact labels, which carry no metric a floor could smooth; the automaton has its own floor, the cell, and a permutation of a finite set is periodic and never mixes, so a lifetime statement for it would need an infinite automaton (discrete-substrate note).
- **Palmer's invariant set postulate** places states on a fractal, measure-zero invariant set and counterfactual settings off it (Palmer 2009; read at the level of its title and our memory of the framework). A Lipschitz density cannot live on a measure-zero set, so for that formulation the two premises are in direct conflict. Later versions of invariant set theory, as we recall them, introduce a finest fractal scale; there the comparison becomes quantitative, that scale against $A_0/h$. The superdeterminist position is surveyed in Hossenfelder and Palmer (2020) (metadata).
- **Quantum mechanics** is untouched: its Bell violations come from entangled states, and it assumes independent settings, consistently with the proposition.

## 3. Consequence for STATE

A small addition to the Newton side: for chaotic setting devices a floor on the prepared state makes Bell's measurement dependence decay exponentially after an Ehrenfest-type onset, and the continuum limit $h \to 0$ reopens the loophole. It is a statement about what the floor buys, with the floor assumed; the necessity question is unchanged. Refereed (Fable, REFINE; items applied: connectedness and mixing, product formulation, hypotheses (a)–(c), scope of the floor, citation roles, style).